Individual Assignment 5

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visualise missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```r
# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

```r
# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")
```

```r
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```r
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
```

```r
# filtering join: only include sites occurring in ncos_sites
semi_join(sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~  "West Pond",
```

```
    site == "NDC" ~ "Devereux Creek"
  )) |>
  # setting factor levels for site
  mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

x-axis: site_full y-axis: log_ave_lit geometry to represent average abundance/L: geom_jitter() geometry to represent medians: geom_point()

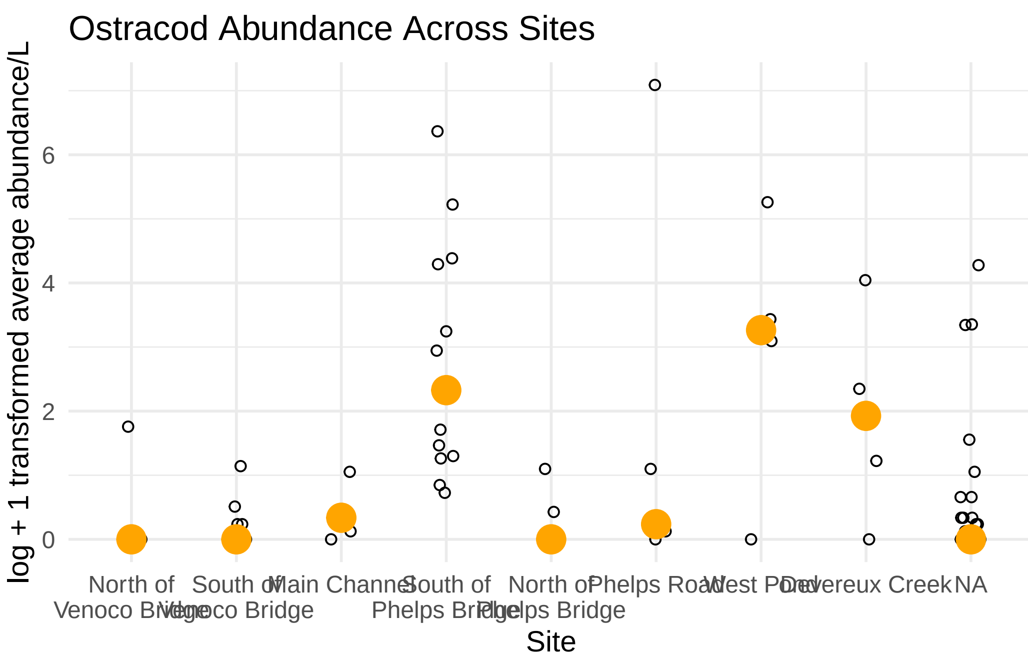
b. Wrangle your data

```
# new object created from clean aquatic inverts
ostracods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "ostracod") |>
  # log + 1 to transform mean abundance/liter
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods)
```

```
# A tibble: 1 x 24
  date_site      date                site  taxon   ave_lit med_ph med_sal med_do
  <chr>          <dttm>              <chr> <chr>     <dbl>  <dbl>   <dbl>  <dbl>
1 2022-05-06_NMC 2022-05-06 00:00:00 NMC   ostrac~       0   9.96    1.45   16.5
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

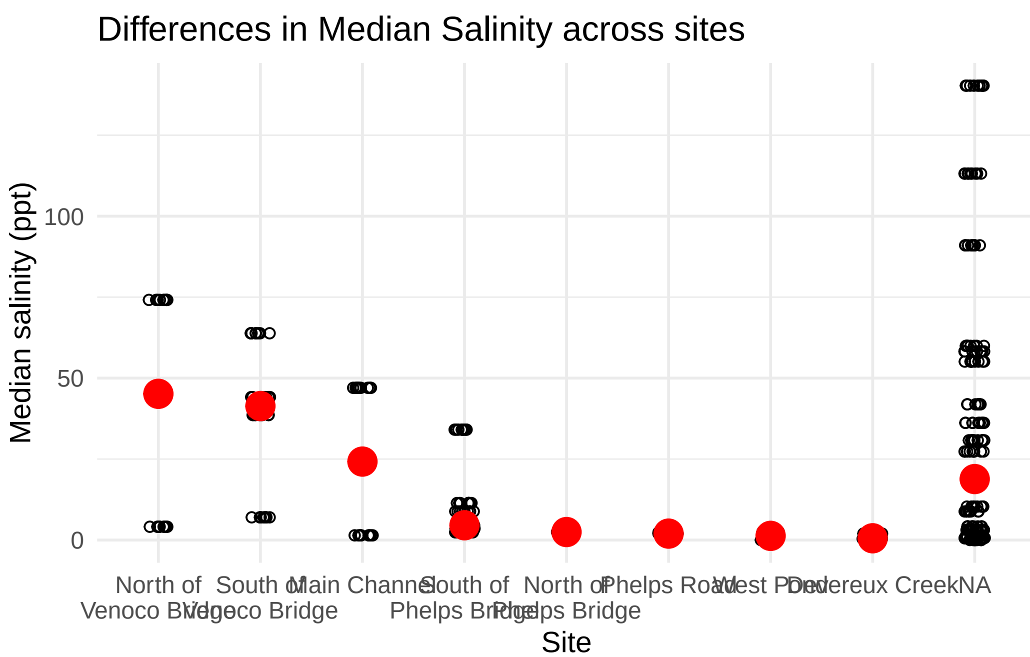
```r
# base layer
ostracodgraph <- ggplot(data = ostracods,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = log_ave_lit)) +
  # salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # points to represent means
  stat_summary(fun = median,
               color = "orange",
               size = 1) +
  # labelling x-axis and y-axis
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracod Abundance Across Sites") +
  # change theme from default
theme_minimal()

# displaying object
ostracodgraph
```



d. Create a multipanel figure

```
# salinity
salinity <- ggplot(data = aq_ins_clean,
       # x-axis: site
       # y-axis: mean salinity
       mapping = aes(x = site_full,
                     y = med_sal)) +
  # salinity measurements at each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # points to represent means
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  # labellingg x-axis and y-axis
  labs(x = "Site",
       y = "Median salinity (ppt)",
```

```
      title = "Differences in Median Salinity across sites") +
  # different theme
theme_minimal()
# displaying object
salinity
```

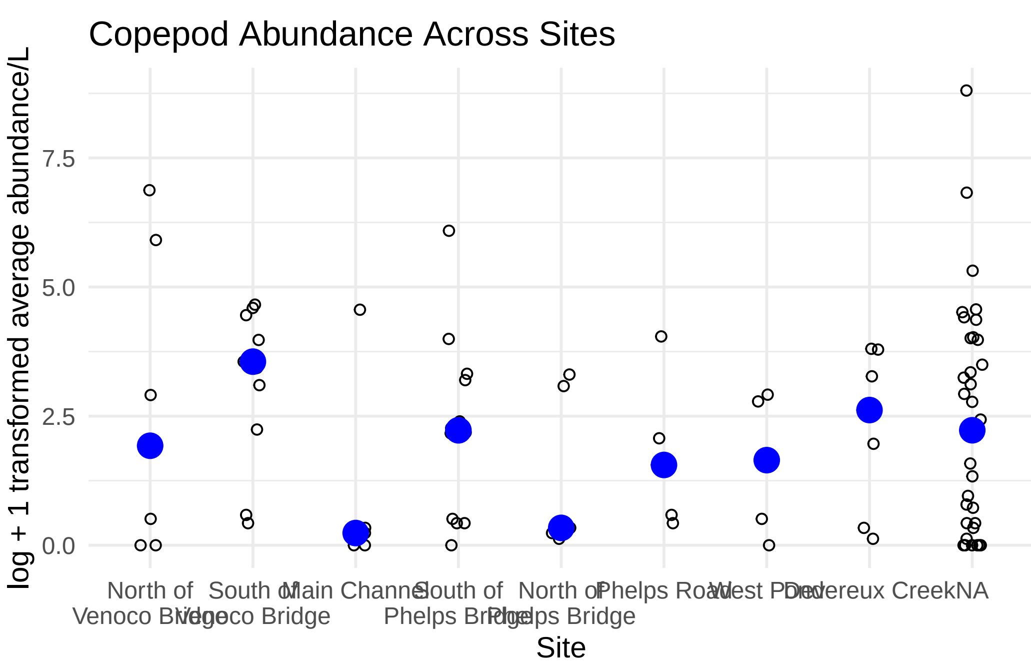



```
# cleaning copepods before graphing
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
8
```

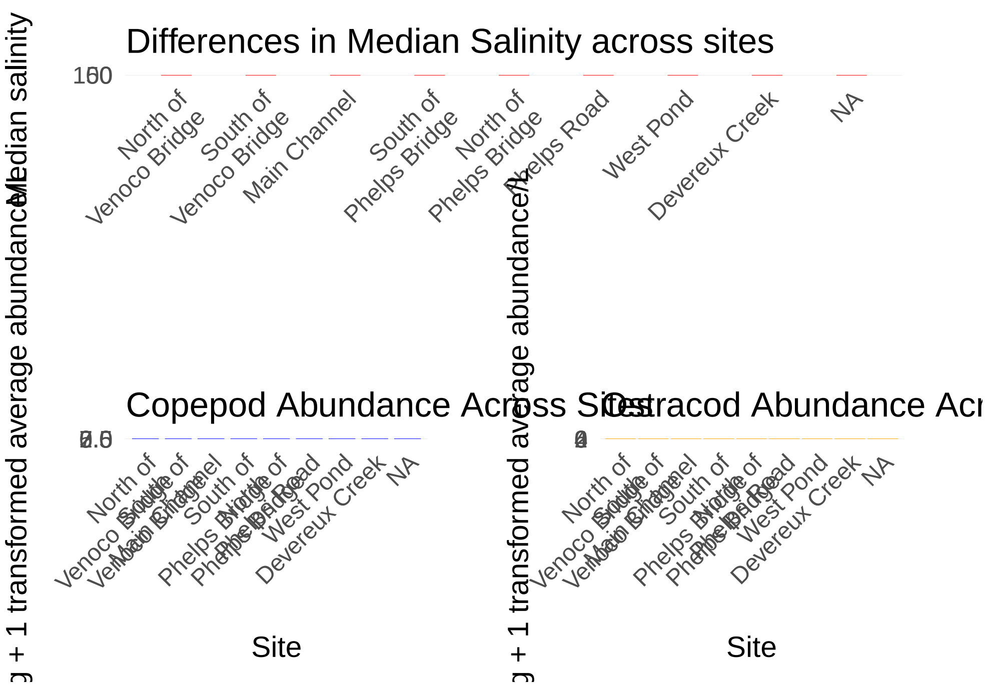
```
[1] 8
```

```
# base layer
copepods <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
```

```
                                  y = log_ave_lit)) +
  # first layer: points to represent observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
               fun = "median",
size = 4,
               color = "blue") +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(15)) +
  # relabelling x- and y-axis, adding title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepod Abundance Across Sites") +
  # different theme
theme_minimal()
# displaying object
copepods
```



```
combined_plot <-
  (salinity + labs(x = NULL)) /
  plot_spacer() /
  (copepods | ostracodgraph) +
  plot_layout(heights = c(1, 0.4, 1)) &
  theme(
    plot.margin = margin(15, 15, 15, 15),
    axis.text.x = element_text(angle = 45, hjust = 1),
    axis.title.x = element_text(margin = margin(t = 15)),
    axis.title.y = element_text(margin = margin(r = 15))
  )

combined_plot
```




```
ggsave("multipanel_figure.png", combined_plot, width = 14, height = 10, dpi = 300)
```

e. Interpret your figure

2. Visualizing total abundance as a function of salinity